Lyudmila Piatkov

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Objective

Electrical Engineering student with hands-on experience in hardware design, embedded systems, and rapid prototyping seeking a Summer 2027 internship in medical devices and surgical robotics, bringing a proven ability to take projects from concept to completion in fast-paced cross-functional teams.

Education

Georgia Institute of Technology | Atlanta, GA *August 2025 - Present*
Bachelor of Science in Electrical Engineering, Spanish Minor, GPA 4.00 Expected Graduation: Fall 2028

Skills

Computer Software: Fusion 360, Blender, Autodesk Inventor, Onshape, Microsoft Office, VSCode, IntelliJ, MATLAB
Programming: Python, Java, C++, HTML, JavaScript
Hardware: Raspberry Pi, Arduino, 3D printers (Prusa, Bambu)
Professional Organizations: Georgia Tech Society of Women Engineers, Georgia Tech Women in ECE
Languages: English (Native), Russian (Fluent), Spanish (Advanced)

Experience

Palitra3D | Wellesley, MA *May 2025 - August 2026*
Consultant and Summer Research Intern
Over 2 summers, assisted a 3D printing company with multiple projects focused on improving quality.

- Engineered end-to-end operational workflows for a 3D-printed topography startup, integrating custom material packaging solutions, web platform design, and market strategies within a core 4-person team to create 50 sample prints
- Simplified 3D-printed color optimization workflow using Muse 3 and Color Passport data to improve color accuracy by 30%

Vertically Integrated Projects | Atlanta, GA *August 2026 - Present*
Mechatronics and Motivation Researcher
Working to design a unique electrical device to quantify and reduce stress in an educational setting.

- Investigated different approaches to analyzing EMG bracelet data using machine learning to quantify signals of stress and anxiety to improve educational outcomes for users

Projects

RESOLV | GT IEEE Robotech Hackathon - 2nd Place Autonomous Track *January 2026*
Built an autonomous robot to inspect and clean dust/sand from solar panels in remote environments (lunar/Martian solar farms) as part of a 4-person team.

- Developed hardware-software integration with a custom 3D-printed chassis, Raspberry Pi, and Streamlit Dashboard that was awarded 2nd place in the Autonomous Track

Robojackets | Battlebots Robots (Conspiraci, Anxietii, Graffiti) *August 2025 - Present*
Designed and built 3lb and 30lb robots to compete at the National Havoc Robot League competitions.

- Conspiraci (3lb undercutter thagomizer) - Designed and built in a team of 7 over the course of 1 year; led assembly of motors, wire harnessing, and battery management
- Anxietii (30lb undercutter spinner) - Improved and iterated on preexisting design for new competition season
- Graffiti (30lb vertical spinner/lifter) - As a sub-team lead, guided design and development of lifter mechanism system

Relevant Coursework

Circuit Analysis: AC/DC circuit theory, nodal/mesh analysis, operational amplifiers, RLC transient response, phasor analysis, and frequency-domain network behaviors
Intro to Signal Processing: Continuous/discrete signals, MATLAB simulation, sampling theorem, FIR filter design, DFT/FFT analysis, and Z-transforms

Leadership or Activities

Robojackets | Battlebots Trip Advisor (“The Foot”) *April 2026 – Present*

- Coordinate all competition travel logistics for 20+ people for 4+ competitions, aligning with Bot leads and leadership on trip planning and deadlines